

C Programming – Summer Training

Assignment 3 Submission

Programs 1 – 15

Contents

- 1 Task 3.1 – WAP in C to generate Fibonacci series up to a given number
- 2 Task 3.2 – WAP in C to reverse the digits of a given number
- 3 Task 3.3 – WAP in C to make a multiplication table of a given number
- 4 Task 3.4 – WAP in C to find the factorial of a given number
- 5 Task 3.5 – WAP in C to find the sum of the digits of a given number
- 6 Task 3.6 – WAP in C to check whether a number is an Armstrong number or not
- 7 Task 3.7 – WAP in C to check whether a given number is a palindrome or not
- 8 Task 3.8 – WAP in C to check whether a given number is prime or not
- 9 Task 3.9 – WAP in C to print all alphabets from a to z
- 10 Task 3.10 – WAP in C to print even numbers up to a given number
- 11 Task 3.11 – WAP in C to display a pattern (left-aligned triangle of stars)
- 12 Task 3.12 – WAP in C to display a pattern (inverted triangle of stars)
- 13 Task 3.13 – WAP in C to display a pattern (increasing number triangle)
- 14 Task 3.14 – WAP in C to display a pattern (decreasing number triangle)
- 15 Task 3.15 – WAP in C to display a pattern (centered pyramid of stars)

Project 1 — Task 3.1

WAP in C to generate Fibonacci series up to a given number

Source Code

```
// Task 3.1 WAP in C to generate Fibonacci series up to given number

#include <stdio.h>

void main() {
    int n, a = 0, b = 1, c;

    printf("Enter the limit: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");

    if (n >= 0)
        printf("%d ", a);

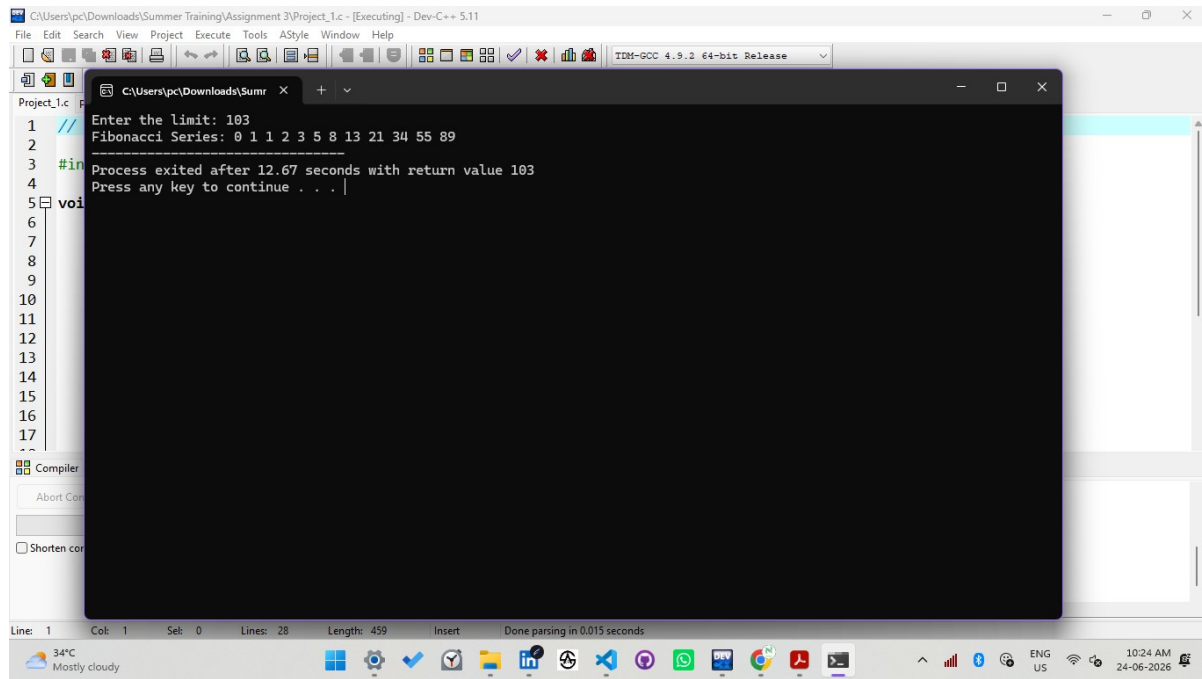
    if (n >= 1)
        printf("%d ", b);

    c = a + b;

    while (c <= n) {
        printf("%d ", c);

        a = b;
        b = c;
        c = a + b;
    }
}
```

Output



The screenshot shows the Dev-C++ IDE with a C++ project named "Project_1.c" open. The code in the editor is as follows:

```
1 // Enter the Limit: 103
2 Fibonacci Series: 0 1 1 2 3 5 8 13 21 34 55 89
3 #include <iostream>
4 using namespace std;
5 void main()
6 {
7     int limit = 103;
8     int a = 0, b = 1, c = 1;
9     cout << "Fibonacci Series: ";
10    while (c <= limit)
11    {
12        cout << c << " ";
13        a = b;
14        b = c;
15        c = a + b;
16    }
17 }
```

The output window shows the following text:

```
Enter the Limit: 103
Fibonacci Series: 0 1 1 2 3 5 8 13 21 34 55 89
Process exited after 12.67 seconds with return value 103
Press any key to continue . . .
```

The status bar at the bottom indicates the file is "Project_1.c", line 1, column 1, and the compiler is "GCC 4.9.2 64-bit Release". The system tray shows the date and time as 10:24 AM on 24-06-2026.

Project 2 — Task 3.2

WAP in C to reverse the digits of a given number

Source Code

```
// Task 3.2 WAP in C to reverse the digit of a given number.

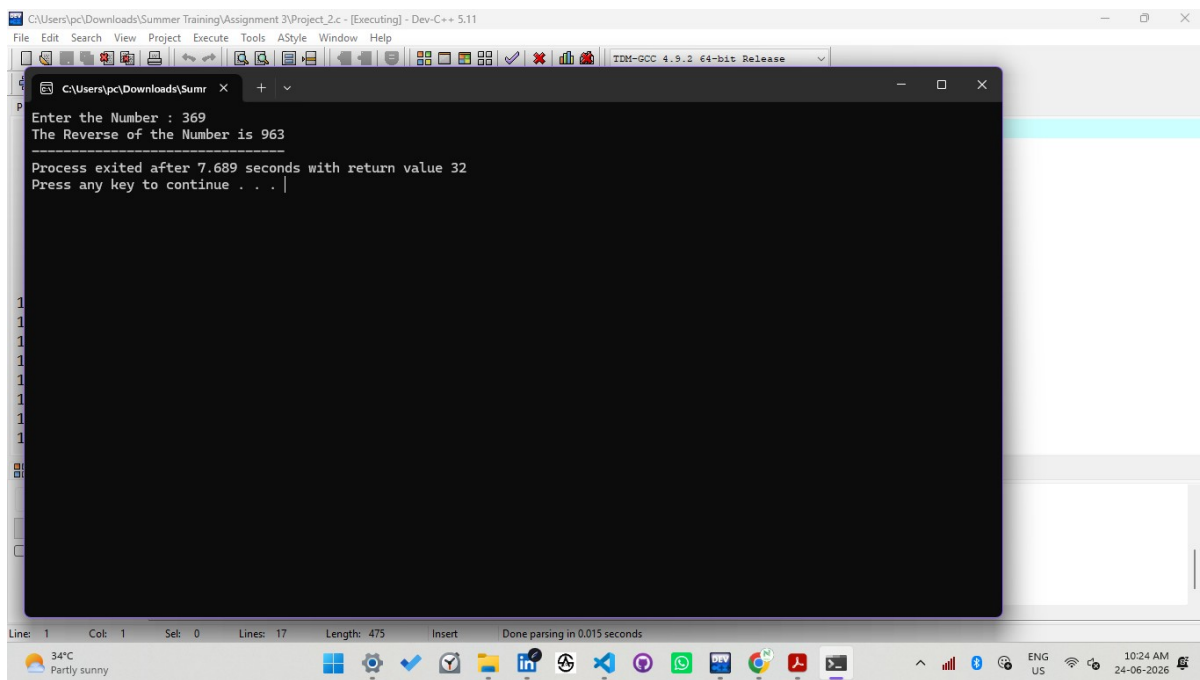
#include<stdio.h>
void main(){
    int num, reverse = 0, digit;

    printf("Enter the Number : ");
    scanf("%d", &num);

    while(num != 0){
        digit = num % 10;          // extract last digit
        reverse = reverse * 10 + digit; // build reversed number
        num /= 10;                 // remove last digit
    }

    printf("The Reverse of the Number is %d", reverse);
}
```

Output



The screenshot shows a Dev-C++ IDE window titled "C:\Users\pc\Downloads\Summer Training\Assignment 3\Project_2.c - [Executing] - Dev-C++ 5.11". The console window displays the following output:

```
Enter the Number : 369
The Reverse of the Number is 963

-----
Process exited after 7.689 seconds with return value 32
Press any key to continue . . . |
```

The status bar at the bottom indicates "Line: 1", "Col: 1", "Sel: 0", "Lines: 17", "Length: 475", and "Done parsing in 0.015 seconds". The system tray shows a temperature of 34°C, "Partly sunny", and the date/time "10:24 AM 24-06-2026".

Project 3 — Task 3.3

WAP in C to make a multiplication table of a given number

Source Code

//Task 3.3 WAP in C to make a table of a given number

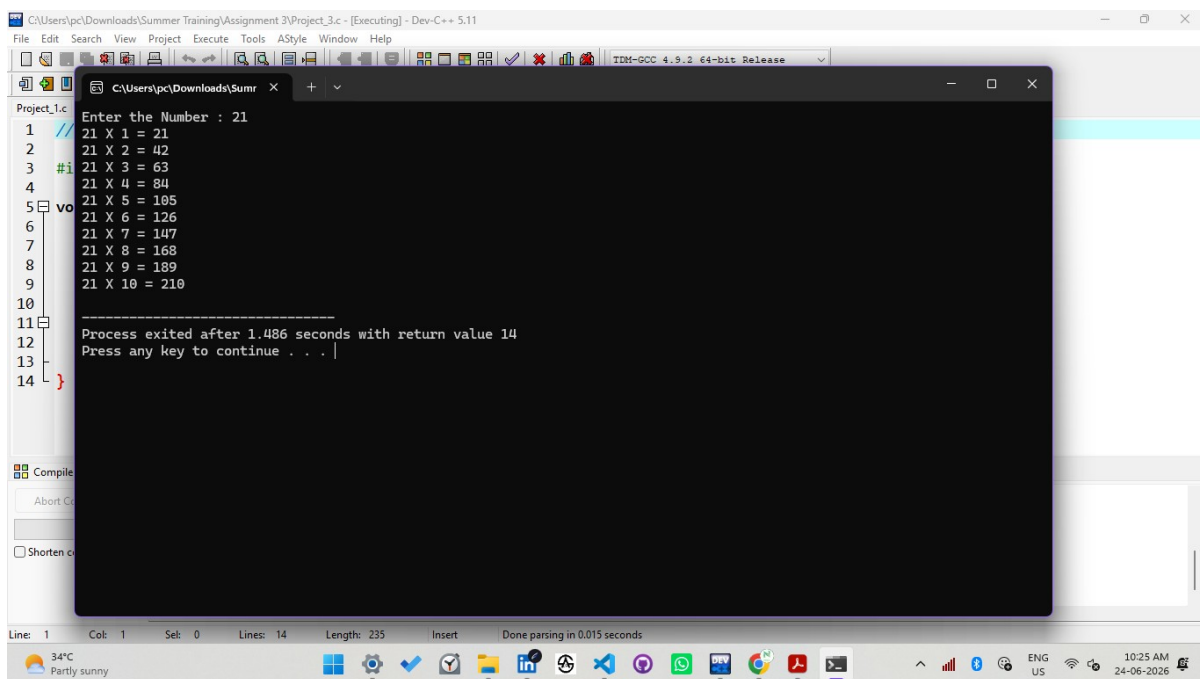
```
#include<stdio.h>

void main(){
    int num,i;

    printf("Enter the Number : ");
    scanf("%d",&num);

    for(i=1;i<=10;i++){
        printf("%d X %d = %d\n",num,i,num*i);
    }
}
```

Output



```
Project_1.c
1 //
2
3 #i
4
5 vo
6
7
8
9
10
11
12
13 }
14

Compile
Abort C
Shorten c

Line: 1 Col: 1 Sel: 0 Lines: 14 Length: 235 Insert Done parsing in 0.015 seconds
34°C Partly sunny 10:25 AM 24-06-2026
```

Enter the Number : 21

```
21 X 1 = 21
21 X 2 = 42
21 X 3 = 63
21 X 4 = 84
21 X 5 = 105
21 X 6 = 126
21 X 7 = 147
21 X 8 = 168
21 X 9 = 189
21 X 10 = 210
```

Process exited after 1.486 seconds with return value 14
Press any key to continue . . .

Project 4 — Task 3.4

WAP in C to find the factorial of a given number

Source Code

```
// Task 3.4 WAP in C to find the factorial of a given number

#include<stdio.h>

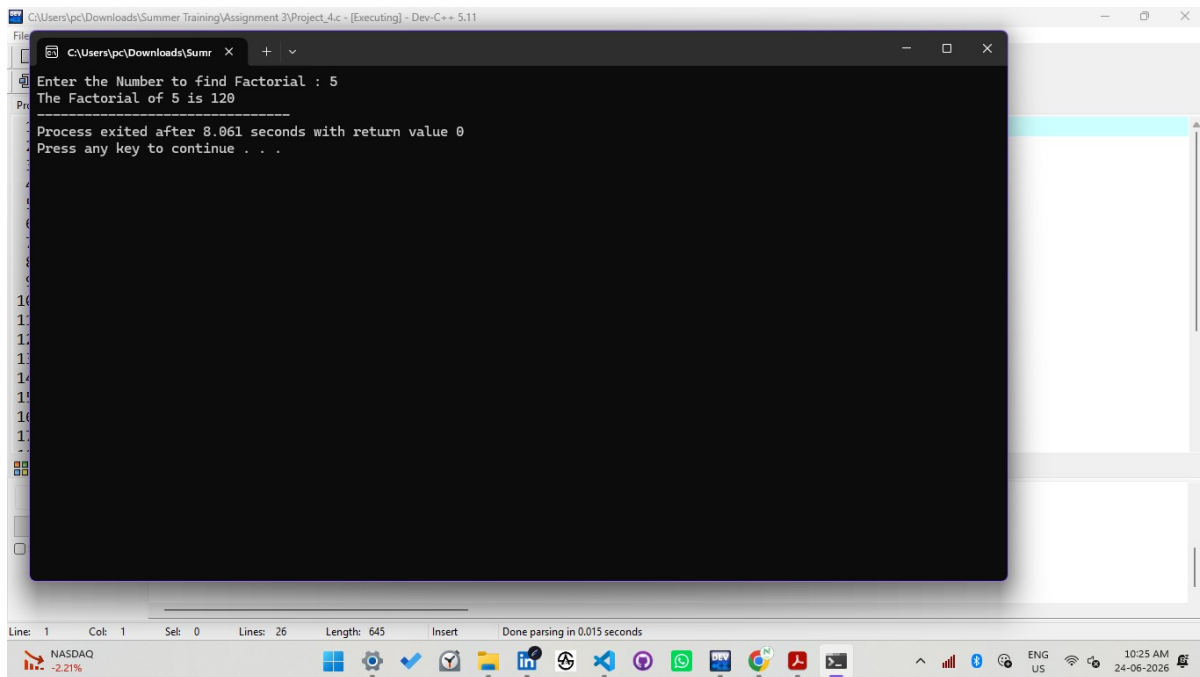
int main(){
    int num, i;
    long long factorial = 1; // use long long, factorials grow fast

    printf("Enter the Number to find Factorial : ");
    scanf("%d", &num);

    if(num == 0 || num == 1){
        printf("The Factorial of the number is 1");
    }
    else if(num < 0){
        printf("Factorial is not defined for negative numbers");
    }
    else{
        for(i = 2; i <= num; i++){
            factorial = factorial * i;
        }
        printf("The Factorial of %d is %lld", num, factorial);
    }

    return 0;
}
```

Output



The screenshot shows a Windows Dev-C++ IDE window titled "C:\Users\pc\Downloads\Summer Training\Assignment 3\Project_4.c - [Executing] - Dev-C++ 5.11". The main window displays the output of the program. The first line shows the prompt "Enter the Number to find Factorial : 5". The second line shows the result "The Factorial of 5 is 120". Below this, a message states "Process exited after 8.061 seconds with return value 0" and "Press any key to continue . . .". The IDE's status bar at the bottom indicates "Line: 1 Col: 1 Sel: 0 Lines: 26 Length: 645 Insert Done parsing in 0.015 seconds". The Windows taskbar at the bottom shows the time as 10:25 AM on 24-06-2026, along with various system icons and the NASDAQ index at -2.21%.

Project 5 — Task 3.5

WAP in C to find the sum of the digits of a given number

Source Code

```
// Task 3.5 WAP in C to find sum of the digit of the given number

#include<stdio.h>

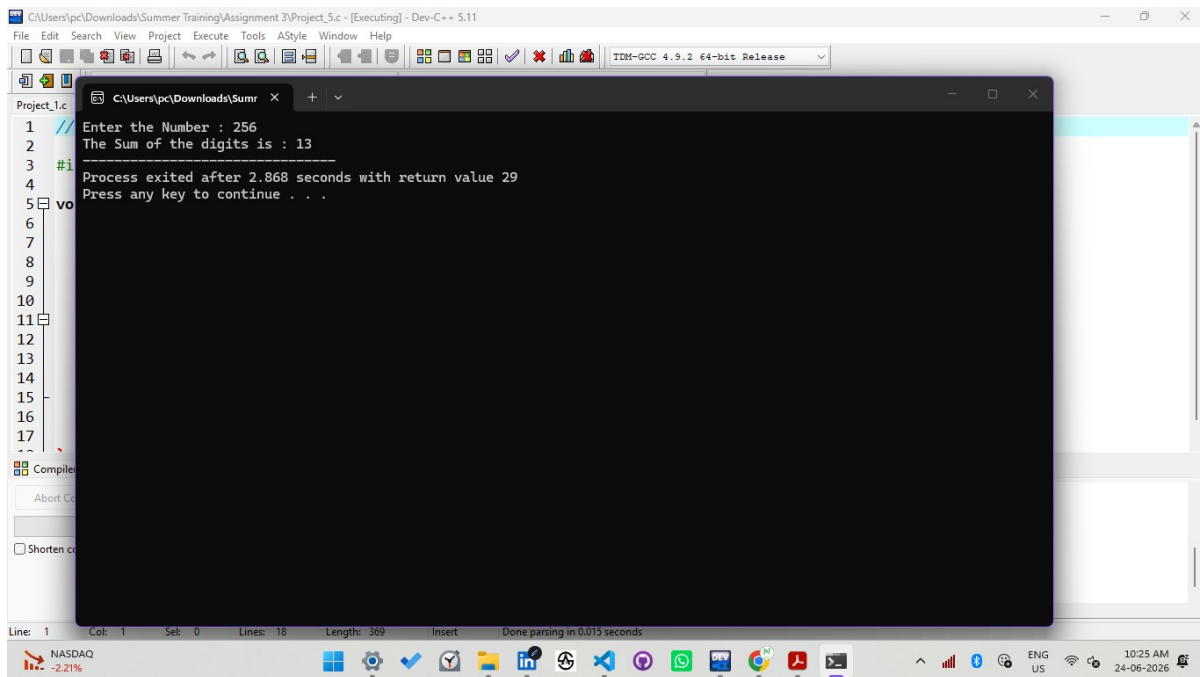
void main(){
    int num, sum = 0, remainder;

    printf("Enter the Number : ");
    scanf("%d", &num);

    while(num > 0){
        remainder = num % 10;
        sum += remainder;
        num = num / 10;
    }

    printf("The Sum of the digits is : %d", sum);
}
```

Output



```
C:\Users\pc\Downloads\Summer Training\Assignment 3\Project_5.c - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TIM-GCC 4.9.2 64-bit Release
C:\Users\pc\Downloads\Sumr x + v
1 //
2 Enter the Number : 256
3 The Sum of the digits is : 13
4 #1
5 Process exited after 2.868 seconds with return value 29
6 Press any key to continue . . .
7
8
9
10
11
12
13
14
15
16
17
18
Line: 1 Col: 1 Sel: 0 Lines: 18 Length: 369 Insert Done parsing in 0.015 seconds
NASDAQ -2.21%
10:25 AM 24-06-2026
```


Project 6 — Task 3.6

WAP in C to check whether a number is an Armstrong number or not

Source Code

```
// Task 3.6 WAP in C to check Armstrong Number or not

#include <stdio.h>

int main(void) {
    int num, total_digits = 0, original, rem, sum = 0, power, temp, i;

    printf("Enter the Number : ");
    scanf("%d", &num);

    if (num < 0) {
        printf("Armstrong check is not valid for negative numbers\n");
        return 0;
    }

    original = num;
    temp = num;

    // Step 1: count total digits
    if (temp == 0) {
        total_digits = 1; // 0 itself has 1 digit
    } else {
        while (temp != 0) {
            temp /= 10;
            total_digits++;
        }
    }

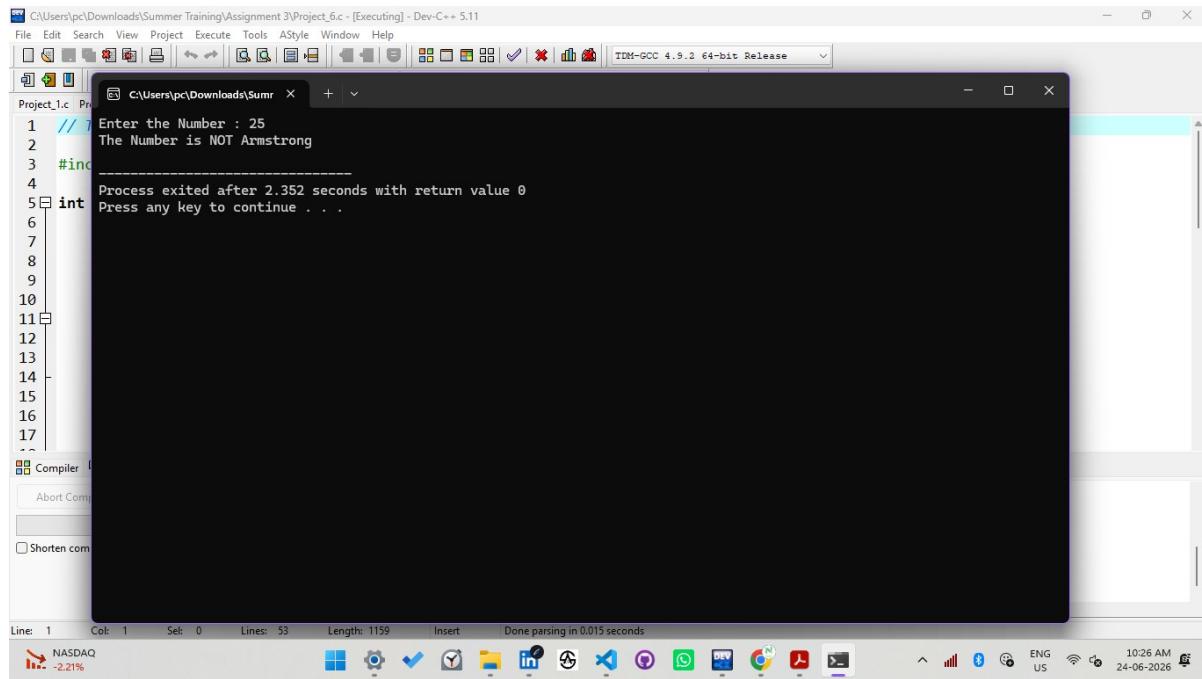
    // Step 2: sum of (each digit ^ total_digits)
    temp = num;
    while (temp != 0) {
        rem = temp % 10;
        power = 1;
        for (i; i < total_digits; i++) {
            power *= rem;
        }
        sum += power;
        temp /= 10;
    }

    // Special case: 0 is trivially Armstrong
    if (original == 0) {
        sum = 0;
    }

    if (sum == original) {
        printf("The Number is Armstrong\n");
    } else {
        printf("The Number is NOT Armstrong\n");
    }

    return 0;
}
```

Output



```
1 // Enter the Number : 25
2 The Number is NOT Armstrong
3 #include <iostream>
4 using namespace std;
5 int main()
6 {
7     int n;
8     cin >> n;
9     // Check if the number is Armstrong or not
10    int sum = 0;
11    int temp = n;
12    while(temp > 0)
13    {
14        int digit = temp % 10;
15        sum += digit * digit * digit;
16        temp /= 10;
17    }
18    if (sum == n)
19        cout << "The Number is Armstrong\n";
20    else
21        cout << "The Number is NOT Armstrong\n";
22    return 0;
23 }
```

Process exited after 2.352 seconds with return value 0
Press any key to continue . . .

Compiler Output: Done parsing in 0.015 seconds

NASDAQ -2.21%

10:26 AM 24-06-2026

Project 7 — Task 3.7

WAP in C to check whether a given number is a palindrome or not

Source Code

```
// Task 3.7 WAP in C to check given number is palindrome or not
```

```
#include<stdio.h>
```

```
void main(){
```

```
    int num,i, first_digit,st, last_digit,original, rem, temp;
```

```
    printf("Enter the Number : ");
```

```
    scanf("%d",&num);
```

```
    st=num;
```

```
    original=num;
```

```
    while(st>0){
```

```
        rem=st%10;
```

```
        temp=rem;
```

```
        st /= 10;
```

```
        first_digit=temp;
```

```
    }
```

```
    last_digit = original%10;
```

```
    if(first_digit == last_digit){
```

```
        printf("The Number is Palindrome");
```

```
    }
```

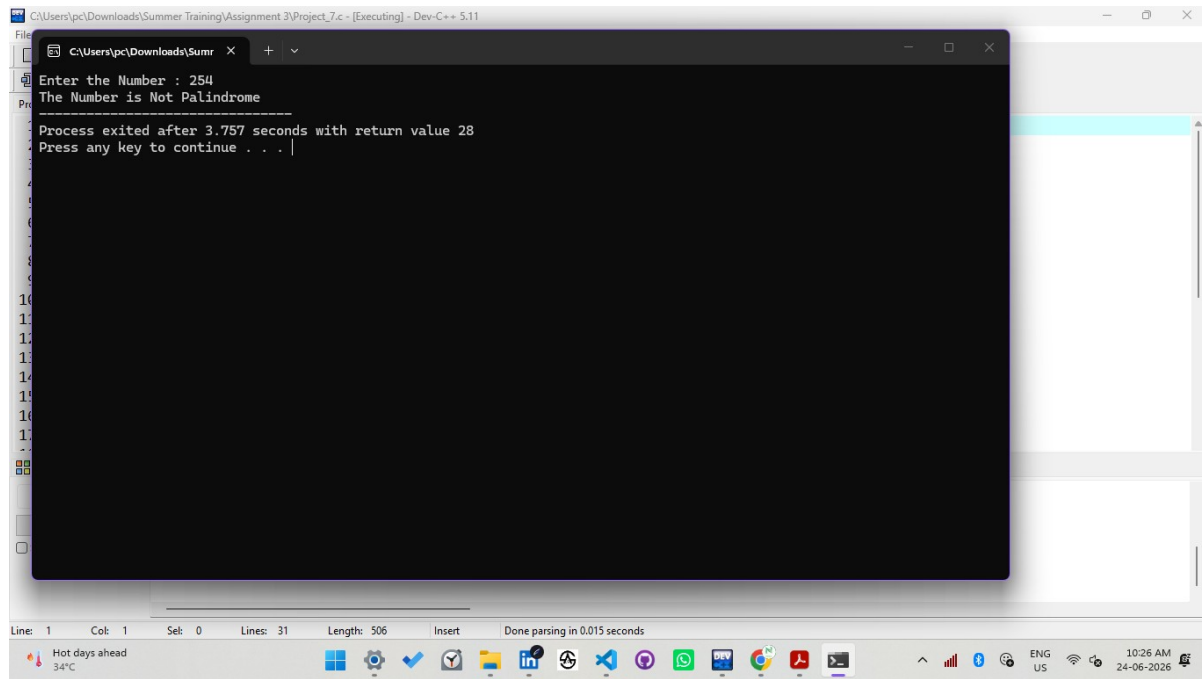
```
    else{
```

```
        printf("The Number is Not Palindrome");
```

```
    }
```

```
}
```

Output



```
C:\Users\pc\Downloads\Summer Training\Assignment 3\Project_7.c - [Executing] - Dev-C++ 5.11
C:\Users\pc\Downloads\Sumr x + v
Enter the Number : 254
The Number is Not Palindrome
-----
Process exited after 3.757 seconds with return value 28
Press any key to continue . . . |
```

Line: 1 Col: 1 Sel: 0 Lines: 31 Length: 506 Insert Done parsing in 0.015 seconds

Hot days ahead 34°C

10:26 AM 24-06-2026

Project 8 — Task 3.8

WAP in C to check whether a given number is prime or not

Source Code

// Task 3.8 WAP in to check given number is prime or not

```
#include<stdio.h>

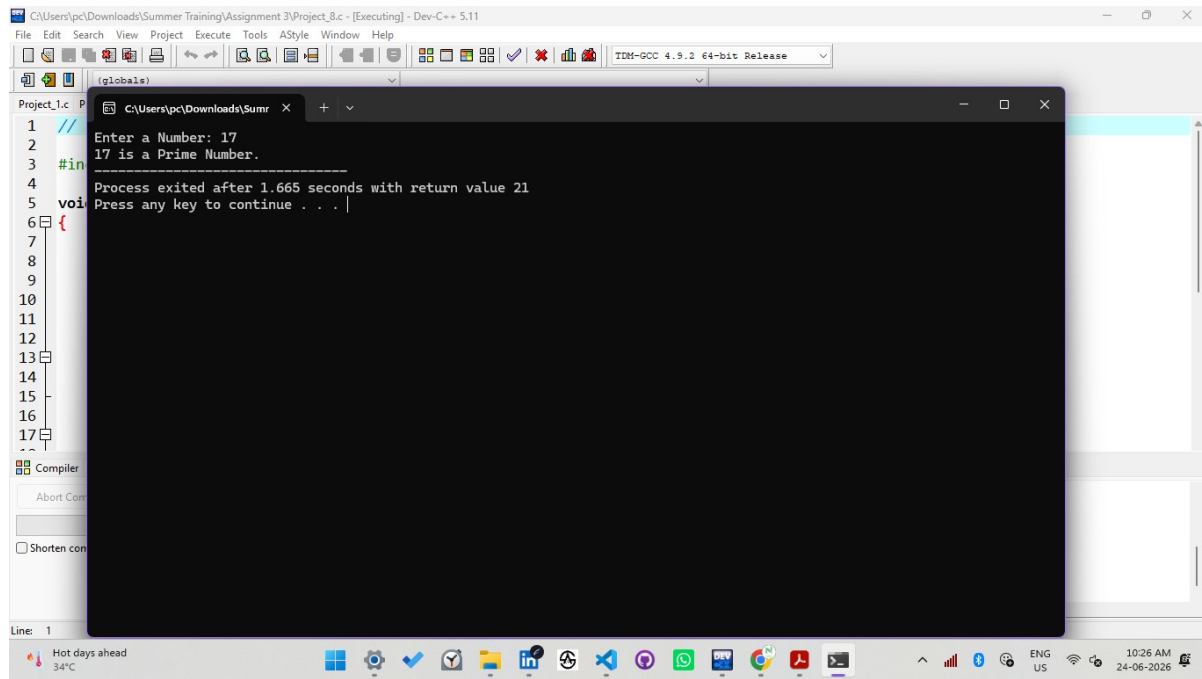
void main()
{
    int num, i, flag = 0;

    printf("Enter a Number: ");
    scanf("%d", &num);

    if(num <= 1)
    {
        printf("%d is not a Prime Number.", num);
    }
    else
    {
        for(i = 2; i < num; i++)
        {
            if(num % i == 0)
            {
                flag = 1;
                break;
            }
        }

        if(flag == 0)
            printf("%d is a Prime Number.", num);
        else
            printf("%d is not a Prime Number.", num);
    }
}
```

Output



The screenshot shows the Dev-C++ IDE with a C++ program being executed. The program prompts the user to enter a number, checks if it is a prime number, and displays the result. The output window shows the execution details, including the return value and the time taken to execute.

```
1 //  
2 Enter a Number: 17  
3 17 is a Prime Number.  
4 -----  
5 Process exited after 1.665 seconds with return value 21  
6 Press any key to continue . . . |  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
..
```

Compiler
Abort Com
Shorten com

Line: 1

Hot days ahead
34°C

10:26 AM
24-06-2026

Project 9 — Task 3.9

WAP in C to print all alphabets from a to z

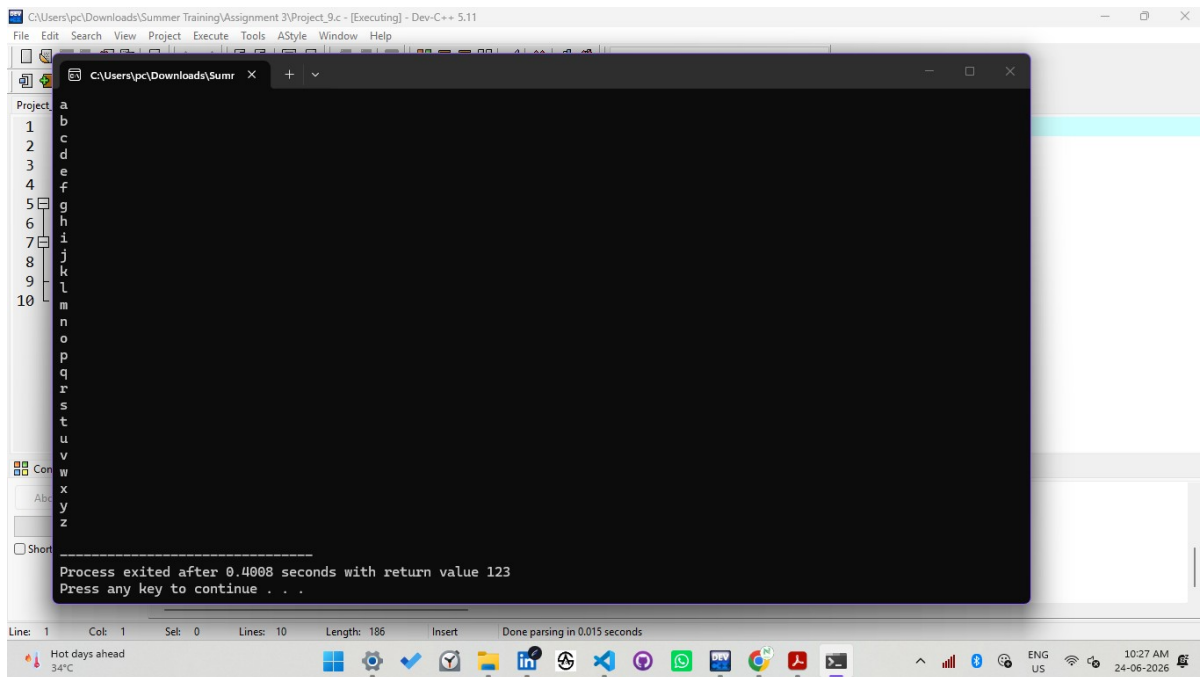
Source Code

```
//Task 3.9 WAP in C to print all alphaet from a to z

#include <stdio.h>

void main() {
    char ch;
    for (ch = 'a'; ch <= 'z'; ch++) {
        printf("%c \n", ch);
    }
}
```

Output



Project 10 — Task 3.10

WAP in C to print even numbers up to a given number

Source Code

```
//Task 3.10 WAP in C to print Even number up to given number.

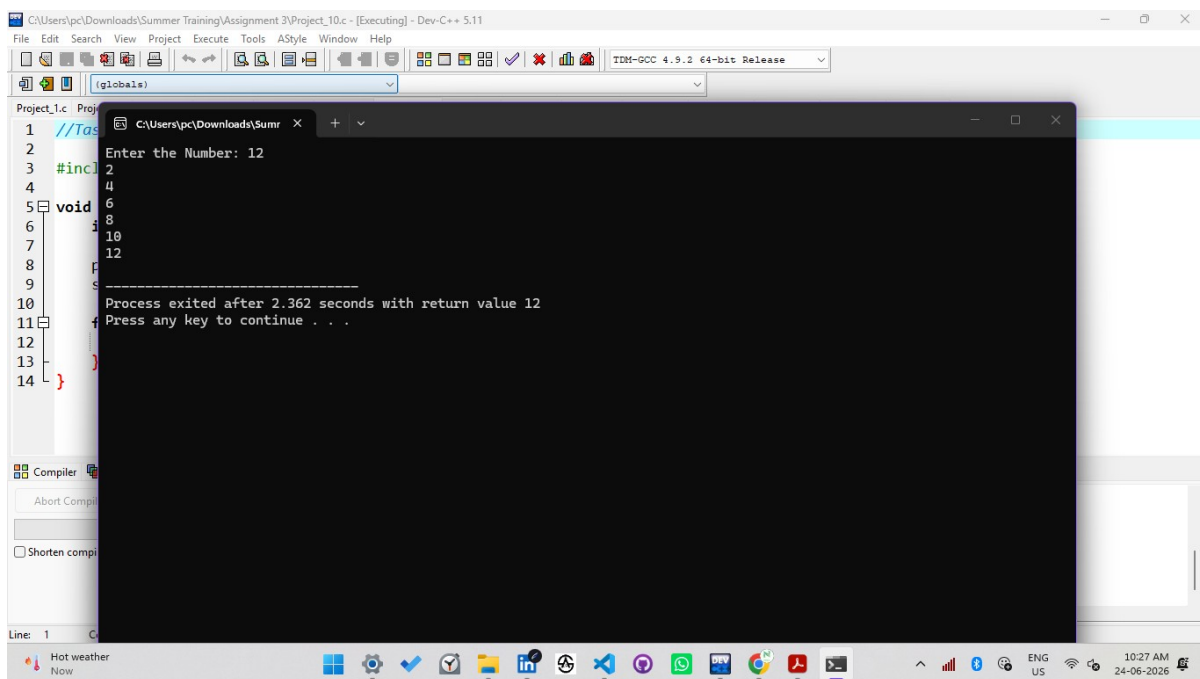
#include<stdio.h>

void main(){
    int num,i=2;

    printf("Enter the Number: ");
    scanf("%d",&num);

    for(i ; i<=num ; i+=2){
        printf("%d \n",i);
    }
}
```

Output



Project 11 — Task 3.11

WAP in C to display a pattern (left-aligned triangle of stars)

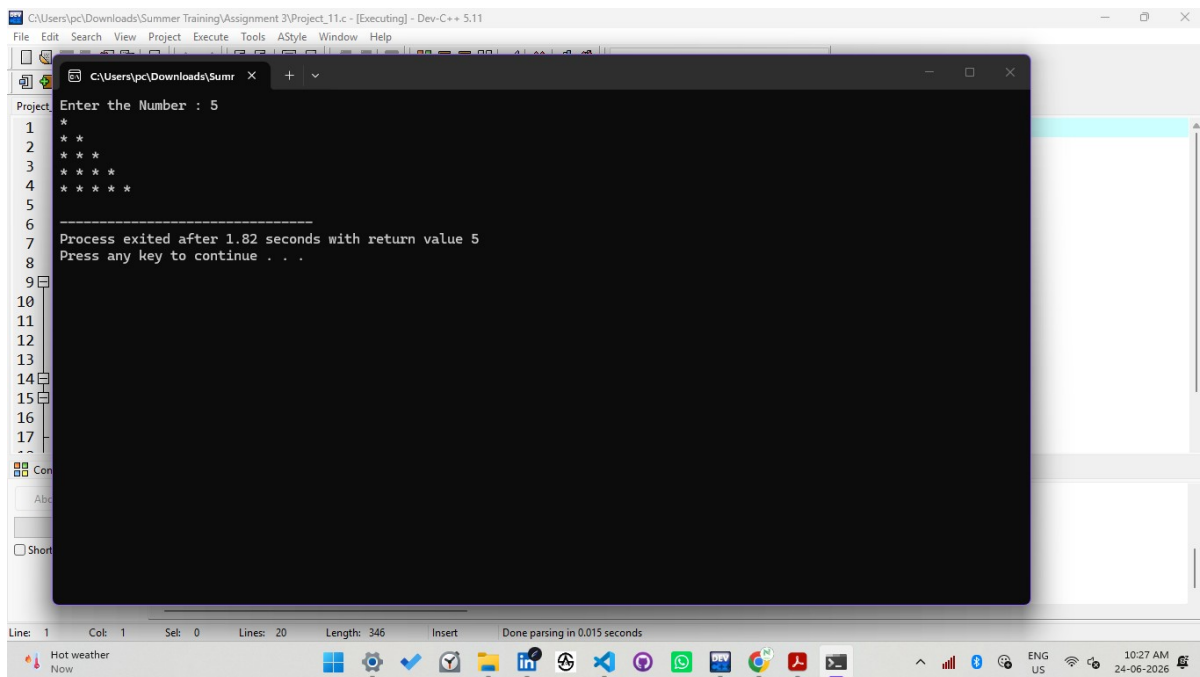
Source Code

```
// Task 3.11 WAP in C to display pattern
/**
/** *
/** * *
/** * * *
/** * * * *
/** * * * * *

#include<stdio.h>
void main() {
    int num, i, j;
    printf("Enter the Number : ");
    scanf("%d", &num);

    for (i = 1; i <= num; i++) {
        for (j = 1; j <= i; j++) {
            printf("* ");
        }
        printf("\n");
    }
}
```

Output



```
C:\Users\pc\Downloads\Summer Training\Assignment 3\Project_11.c - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
C:\Users\pc\Downloads\Sumr x + v
Project: Enter the Number : 5
1 *
2 * *
3 * * *
4 * * * *
5 * * * * *
6 -----
7 Process exited after 1.82 seconds with return value 5
8 Press any key to continue . . .
9
10
11
12
13
14
15
16
17
Con
Abn
Short
Line: 1 Col: 1 Sel: 0 Lines: 20 Length: 346 Insert Done parsing in 0.015 seconds
Hot weather Now
24-06-2026 10:27 AM
```

Project 12 — Task 3.12

WAP in C to display a pattern (inverted triangle of stars)

Source Code

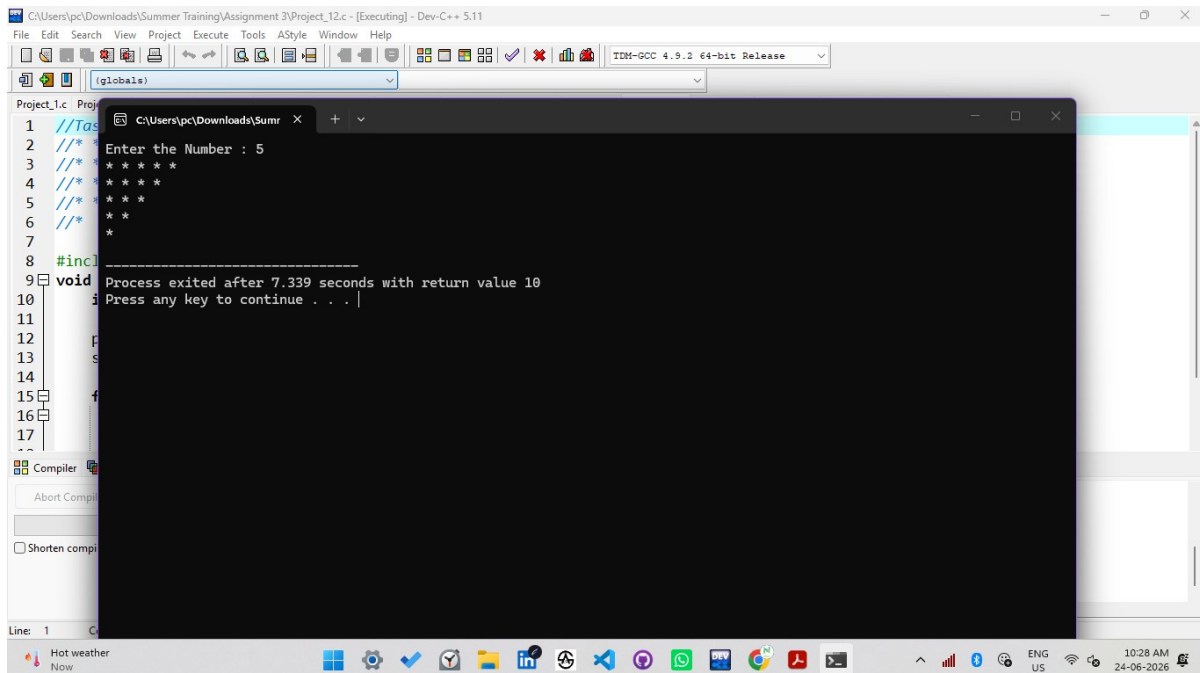
```
//Task 3.12 WAP in C to display pattern
/* * * * * *
/* * * * *
/* * * *
/* * *
/* *
/*
/*

#include<stdio.h>
void main(){
    int num, i, j;

    printf("Enter the Number : ");
    scanf("%d", &num);

    for (i = num; i >= 1; i--) { // start from num, decrease to 1
        for (j = 1; j <= i; j++) {
            printf("* ");
        }
        printf("\n");
    }
}
```

Output



Project 13 — Task 3.13

WAP in C to display a pattern (increasing number triangle)

Source Code

```
// Task 3.13 WAP in C to display pattern
//1
//1 2
//1 2 3
//1 2 3 4
//1 2 3 4 5

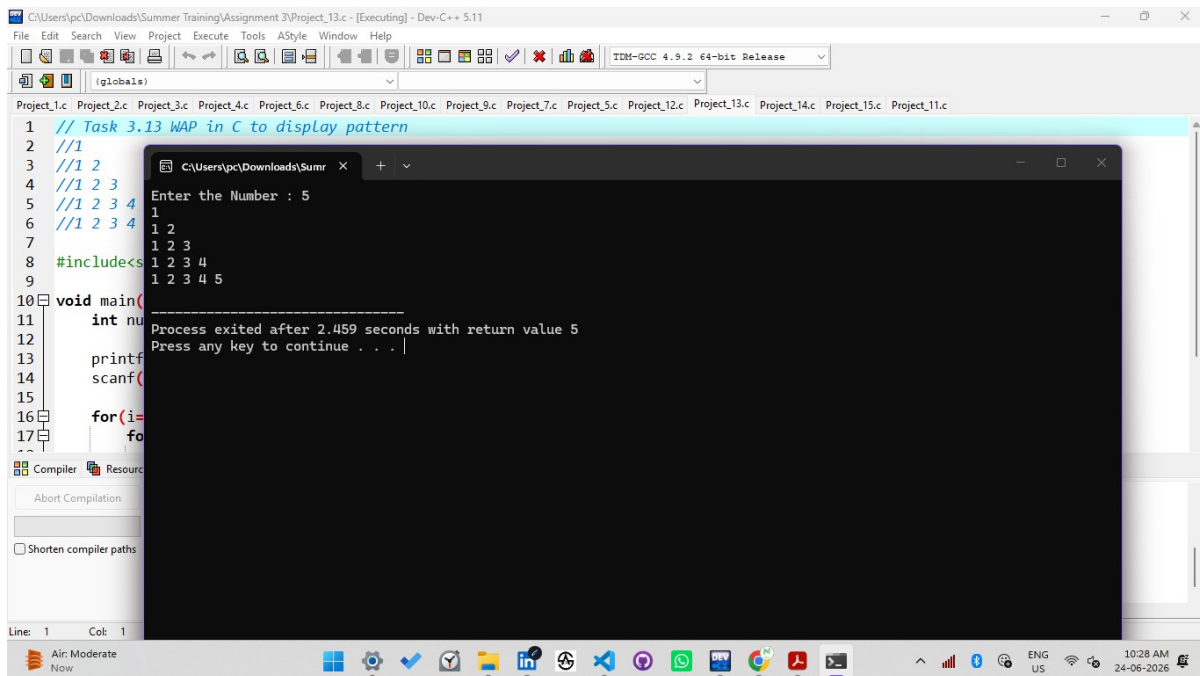
#include<stdio.h>

void main(){
    int num,i,j;

    printf("Enter the Number : ");
    scanf("%d",&num);

    for(i=1 ; i<=num ; i++){
        for(j=1 ; j <=i ; j++){
            printf("%d ",j);
        }
        printf("\n");
    }
}
```

Output



```
1 // Task 3.13 WAP in C to display pattern
2 //1
3 //1 2
4 //1 2 3
5 //1 2 3 4
6 //1 2 3 4 5
7
8 #include<stdio.h>
9
10 void main()
11 {
12     int num;
13     printf("Enter the Number : ");
14     scanf("%d",&num);
15
16     for(i=1 ; i<=num ; i++){
17         for(j=1 ; j <=i ; j++){
18             printf("%d ",j);
19         }
20         printf("\n");
21     }
22 }
```

Process exited after 2.459 seconds with return value 5
Press any key to continue . . .

Project 14 — Task 3.14

WAP in C to display a pattern (decreasing number triangle)

Source Code

```
//Task 3.14 WAP in C to display pattern
//1 2 3 4 5
//1 2 3 4
//1 2 3
//1 2
//1
//1

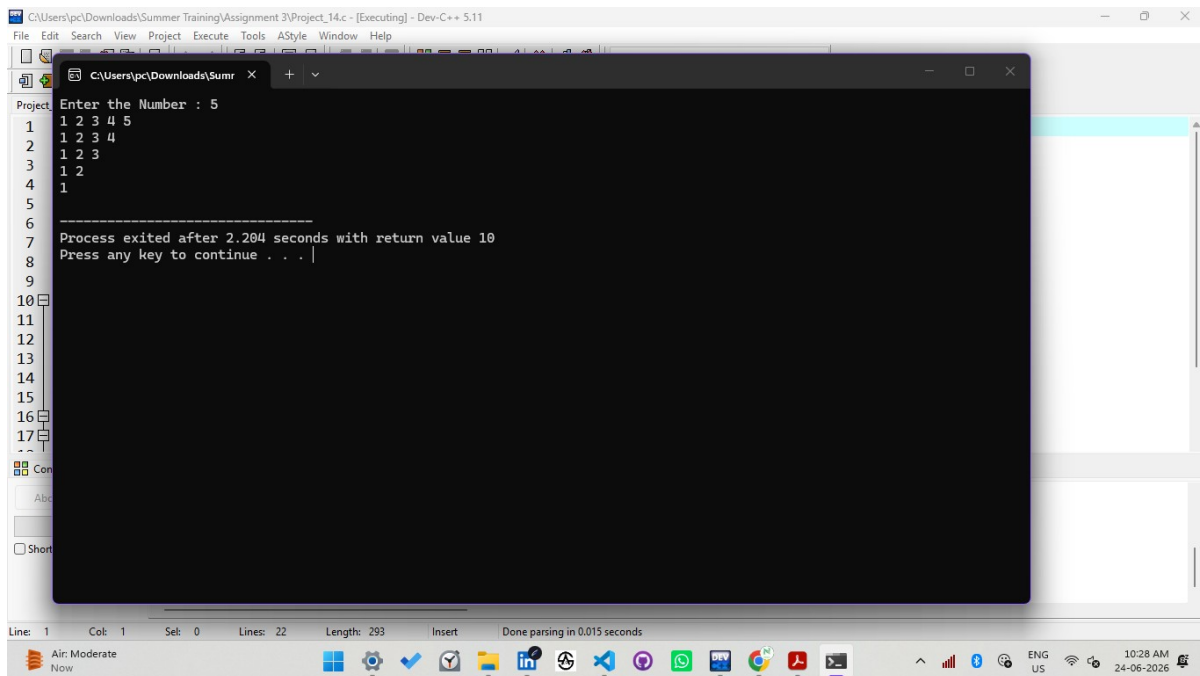
#include<stdio.h>

void main(){
    int num,i,j;

    printf("Enter the Number : ");
    scanf("%d",&num);

    for(i=num;i>=1;i--){
        for(j=1;j<=i;j++){
            printf("%d ",j);
        }
        printf("\n");
    }
}
```

Output



The screenshot shows a C++ IDE window titled "C:\Users\pc\Downloads\Summer Training\Assignment 3\Project_14.c - [Executing] - Dev-C++ 5.11". The code editor displays the source code for the program. The output window shows the execution results: "Enter the Number : 5", followed by the pattern of numbers (1 2 3 4 5, 1 2 3 4, 1 2 3, 1 2, 1) and a message "Process exited after 2.204 seconds with return value 10". The status bar at the bottom indicates "Line: 1 Col: 1 Sel: 0 Lines: 22 Length: 293 Insert Done parsing in 0.015 seconds". The system tray at the bottom shows the date and time as "24-06-2026 10:28 AM".

```
Enter the Number : 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
Process exited after 2.204 seconds with return value 10
Press any key to continue . . .
```

Project 15 — Task 3.15

WAP in C to display a pattern (centered pyramid of stars)

Source Code

```
//Task 3.15 WAP in C to display following pattern
/*
/* * *
/* * * * *
/* * * * * *
/* * * * * *
/* * * * * *

#include<stdio.h>
void main(){
    int num, i, j, k, rows, maxStars;

    printf("Enter the Number : ");
    scanf("%d", &num);

    rows = (num + 1) / 2;          // number of rows
    maxStars = 2 * rows - 1;      // stars in the last (widest) row

    for(i = 1; i <= num; i += 2){
        // print leading spaces for centering
        for(k = 1; k <= (maxStars - i) / 2; k++){
            printf("  ");        // 2 spaces to match "* " width
        }
        // print stars
        for(j = 1; j <= i; j++){
            printf("* ");
        }
        printf("\n");
    }
}
```

Output

The screenshot shows the Dev-C++ IDE with a C++ project named "Project_1.c" open. The code in the editor is as follows:

```
1 //Task 1
2 //Write a program to take an input number and print the star pattern
3 //*
4 // * *
5 // * * *
6 // * * * *
7 // * * * * *
8 #include <stdio.h>
9 void main()
10 {
11     int n;
12     printf("Enter the Number : ");
13     scanf("%d", &n);
14     //Print the star pattern
15     for (int i = 1; i <= n; i++)
16     {
17         for (int j = 1; j <= i; j++)
18             printf("* ");
19         printf("\n");
20     }
21 }
```

The output window shows the execution results:

```
Enter the Number : 5
*
* *
* * *
* * * *
* * * * *
Process exited after 1.874 seconds with return value 5
Press any key to continue . . .
```

The IDE interface includes a menu bar (File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, Help), a toolbar, a compiler selection dropdown (TDM-GCC 4.9.2 64-bit Release), and a status bar at the bottom showing system information (Air: Moderate Now, 10:28 AM, 24-06-2026).